Treebank Creation for a Small Corpus

NLP Project

CS458

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Survey

What is a Treebank?

A treebank, in the field of NLP, is a parsed text corpus which annotates syntactic sentence structure. It revolutionized computational linguistics in the early 1990s.

The name treebank comes from the fact that the structure is commonly represented as a tree structure. It is also known as a parsed corpus.

Survey

Treebanks are usually created on a corpus which is already annotated with part-of-speech tags. They can be created manually or semi-automatically.

Examples of treebanks include:
Penn treebank
Universal Dependencies
Multi-treebank

Treebanks are foundational training data in NLP, which help models to learn structures and relationships between words. This helps for parsing and translation tasks. Treebanks turn raw text into structured data that machines can learn from.

Survey

The Penn treebank is one of the most widely used in the world. It has high-quality manually annotation. The disadvantage is that it is entirely in English and has limited capability for usage in other languages.

Universal Dependencies is a modern treebank framework. It uses dependency parsing and has 100+ languages available. It is widely used in other tools like spaCy.

One disadvantage common to both is that they do not correctly capture domain-specific nuances.

Implementation

Dataset

Cricket report headlines were chosen as a dataset to create the small corpus. Specifically, the headline and 2nd line of ESPNCricinfo match reports of the 74 matches in the IPL 2025 season were taken as data for the dataset.

This resulted in the collection of 148 sentences for the dataset.

Implementation

Preprocessing

The 148 sentences were accessed from ESPNCricinfo and stored into a .csv file with other relevant details like match number and headline/summary

Next, the whitespaces at start and end of sentences were removed for better results. All the sentences were then tokenized using spaCy in python.

Implementation

Annotation

Again using spaCy in python, all of the 148 tokenized sentences were automatically annotated with part-of-speech tags and dependency tags.

Afterwards, 30 sentences (~20% of the dataset) were manually annotated. This subset is sufficiently large to capture key linguistic patterns while keeping the manual correction process manageable.

Implementation

Custom Tags

Custom tags related to the cricket domain were added to the first 30 sentences.

The tags used were:
PLAYER, TEAM, VENUE, TOURNAMENT, RUN_EVENT, WICKET, PARTNERSHIP, BOUNDARY, OVER, SCORE, RESULT, MARGIN, PHASE, AWARD and ROLE

Any words falling outside of these categories were tagged as ‘O’ for outside.

Implementation

Data Format

After performing all the manual corrections and custom tagging for the first 30 sentences of the dataset, the .csv file was converted into a .json file in order to perform better analysis. This was done using two separate python files.

In the .json file the data stored was:
tokens, custom tags, pos tags, dep tags

Analysis

Dataset

The dataset contains 148 sentences.
The average length of a sentence is 15.84 tokens
Comparing the average length based on type:

Headline: 11.43 tokens
Summary: 20.26 tokens

Analysis

Part of Speech (POS)

The distribution of Part of Speech is shown:

The distribution shows that proper nouns (like players and teams) are the most used. It is closely followed by regular nouns like win, run, wicket, game, ball, bat, etc.

NUM appears once per sentence for player score or team score.

We see CCONJ (conjunction) as low due to the simple sentence structure used by headlines and summaries

PROPN: 430
NOUN: 415
ADP: 282
VERB: 270
NUM: 149
ADJ: 149
DET: 130
PUNCT: 128
PART: 98
CCONJ: 82

Analysis

Dependency

The distribution of dependency tags is shown:

The high frequency of prepositional dependencies (prep, pobj) reflects the nature of cricket commentary, where scores, locations, and match situations are often expressed through structured phrases such as “200 for 6” or “win in Chennai”.

ROOT has a count almost equal to number of sentences, thus showing that sentences are mostly single clause.

High AMOD count shows that the sentences are descriptive in nature (quick fifty)

PREP: 261
POBJ: 246
NSUBJ: 199
DOBJ: 191
ROOT: 149
AMOD: 144
PUNCT: 130
DET: 129
COMPOUND: 128
CONJ: 92

Analysis

Manual vs Automatic

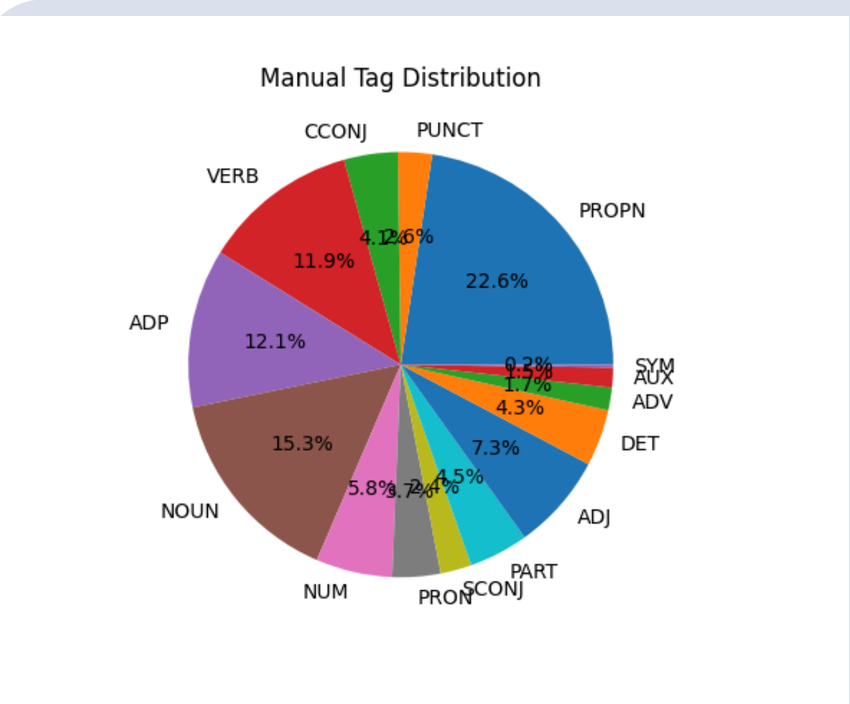
Comparison of Manual and Automatic tagged sentences:

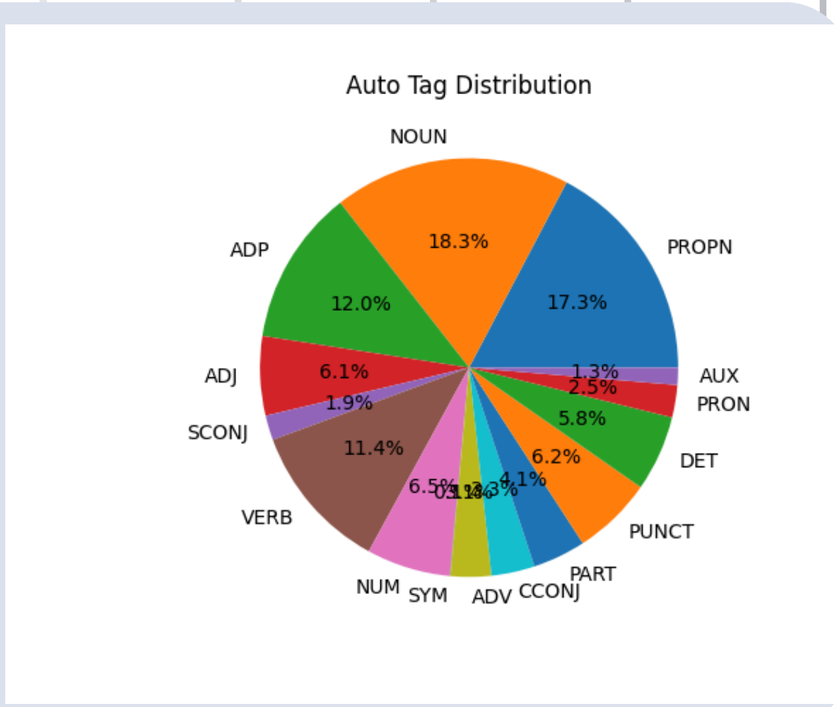
The major difference seen in manual and automatic tagged sentences is the frequency of proper nouns and nouns. In the automatic tagging, many proper nouns like player names and team names were incorrectly classified as regular nouns. Also adjectives were incorrectly classified as adverbs in automatic tagging.

Automatic tagging captures general structure well but lacks precision in domain-specific and entity-heavy contexts.

Manual Tags:	Auto Tags:
PROPN: 22.63	NOUN: 18.29
NOUN: 15.30	PROPN: 17.28
ADP: 12.07	ADP: 12.01
VERB: 11.85	VERB: 11.43
ADJ: 7.33	NUM: 6.49
NUM: 5.82	PUNCT: 6.17
PART: 4.53	ADJ: 6.11
DET: 4.31	DET: 5.85
CCONJ: 4.09	PART: 4.09
PRON: 3.66	CCONJ: 3.35
PUNCT: 2.59	ADV: 3.14
SCONJ: 2.37	PRON: 2.50
ADV: 1.72	SCONJ: 1.91
AUX: 1.51	AUX: 1.33
SYM: 0.22	SYM: 0.05

Analysis


Manual Tag Distribution
CCONJ
PUNCT
VERB
PROPN
11.9%
22.6%
ADP
12.1%
SYM
AUX
ADV
4.3%
DET
15.3%
7.3%
ADJ
NOUN
5.8%
4.5%
PART
NUM
PRON
SCONJ


Auto Tag Distribution
NOUN
18.3%
PROPN
ADP
12.0%
17.3%
ADJ
6.1%
1.3%
AUX
2.5%
PRON
1.9%
5.8%
SCONJ
11.4%
6.2%
DET
4.1%
VERB
PUNCT
PART
NUM
SYM
ADV
CCONJ

Analysis

Headline vs Summary

Comparison of Headline and Summary sentences:

Headlines have a high concentration of proper nouns and adjectives. This indicates a compressed, entity-heavy style for quick information.

In contrast, summaries show higher use of prepositions and numerical tokens. This shows that the slightly deeper details of scores and interactions.

Headlines are concise and entity-dense, while summaries are longer and more contextually descriptive.

POS TAG % (HEADLINE)	POS TAG % (SUMMARY)
PROPN: 27.42%	NOUN: 16.94%
NOUN: 19.03%	PROPN: 13.21%
VERB: 11.94%	ADP: 13.01%
ADP: 10.28%	VERB: 11.27%
ADJ: 6.97%	NUM: 7.67%

Analysis

Custom Tags

This distribution highlights the prime importance of players, teams, results and scores in the report headlines and summaries. This set of tags forms a gap between the other minor set.

The set of Tournament, Wicket, Run, Margin, Role, etc. are not having much importance in the sentences, as they would be explained in detail in the main report article.

Thus it appears that the top 2 sentences show focus on the main details like important players, winning and losing team and the score.

=== CUSTOM TAGS ===

O: 59.91%
PLAYER: 12.93%
TEAM: 8.84%
RESULT: 6.47%
SCORE: 3.02%

TOURNAMENT: 1.94%
WICKET: 1.94%
ROLE: 1.94%
MARGIN: 1.51%
RUN_EVENT: 0.65%
PARTNERSHIP: 0.43%
VENUE: 0.43%

Analysis

Headline vs Summary (Custom Tags)

As seen before, headlines have a high concentration of Player, Result and Team, and very few mentions of other aspects like score, role, margin and venue.

Summary lines also mainly show player and team, but also include more minor details like wickets, margin, role, partnership, etc.

This shows that headlines are more likely to have similar composition most of the time, while summary lines have a varied distribution of details.

=== CUSTOM TAG % (HEADLINE) ===

O: 43.79%
PLAYER: 20.26%
RESULT: 13.07%
TEAM: 13.07%

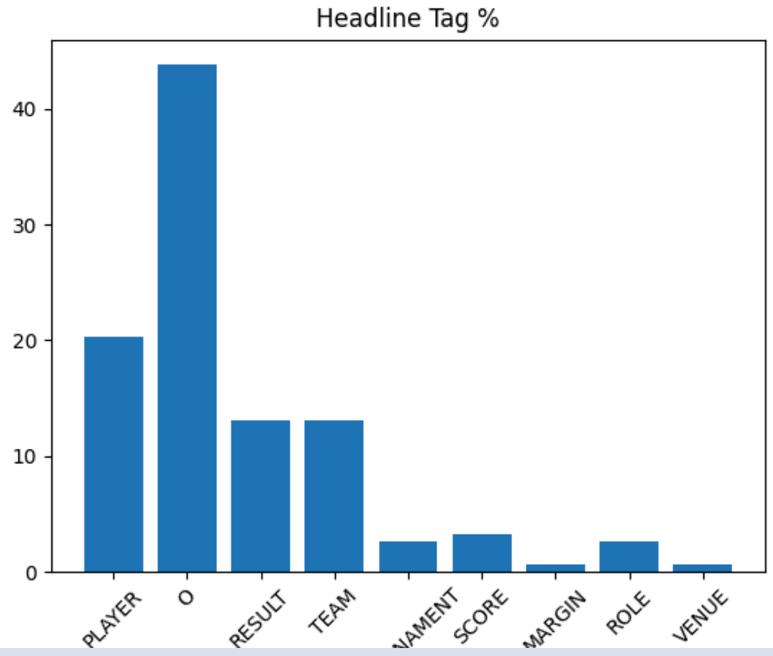
SCORE: 3.27%
TOURNAMENT: 2.61%
ROLE: 2.61%
MARGIN: 0.65%
VENUE: 0.65%

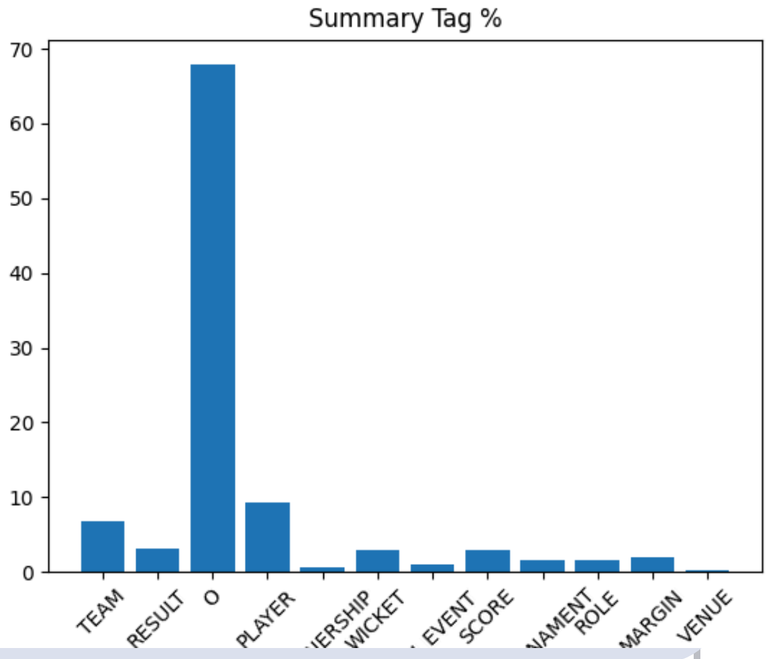
=== CUSTOM TAG % (SUMMARY) ===

O: 67.85%
PLAYER: 9.32%
TEAM: 6.75%

RESULT: 3.22%
WICKET: 2.89%
SCORE: 2.89%
MARGIN: 1.93%
TOURNAMENT: 1.61%
ROLE: 1.61%
RUN_EVENT: 0.96%
PARTNERSHIP: 0.64%
VENUE: 0.32%

Analysis


Headline Tag %
0
10
20
30
40
PLAYER
O
RESULT
TEAM
NAMENT
SCORE
MARGIN
ROLE
VENUE


Summary Tag %
0
10
20
30
40
50
60
70
TEAM
RESULT
O
PLAYER
WICKET
EVENT
SCORE
NAMENT
ROLE
MARGIN
VENUE

Innovation

Domain-specific tagging

The automated tagging does not capture all the nuances and context of specific domains, which is important.

For example, a Player and a Team would both be considered Proper nouns. There would be no difference in number signifying runs and wickets.

This specific context was added using custom tags to show their importance

Innovation

Partial Manual Annotation

30 out of 148 sentences were selected for manual correction in the annotation. This was done so that a comparison could be made between automated and manual annotations

The analysis done on the same reveals the drawbacks in the fully automated tagging for the cricket-domain specific sentences (mistag of proper nouns as nouns)

Conclusion

This project developed a domain-specific treebank using IPL cricket match summaries, combining automatic linguistic annotation with targeted manual corrections.

The analysis demonstrates that while automated methods provide a reasonable baseline, they often lack precision in capturing domain-specific entities and contextual nuances, which are better handled through manual annotation.

Comparisons between headlines and summaries reveal distinct structural patterns, with headlines being concise and entity-heavy, and summaries offering more descriptive and context-rich information.

Overall, the project highlights the importance of domain adaptation in NLP and shows how even a partially annotated treebank can provide valuable linguistic and analytical insights.

Thank You